

ICE 229 Discrete Mathematics

Chapter 9 Relations

9.1 Relations and their properties

- Relationships between elements of sets are represented using the structure called a **relation**
- A subset of Cartesian product of the sets
- Example: a student and his/her ID

Binary relation

- The most direct way to express a relationship between elements of two sets is to use **ordered pairs** made up of two related elements
- **Binary relation:** Let A and B be sets. A binary relation from A to B is a subset of $A \times B$
- A binary relation from A to B is a set R of ordered pairs where the 1st element comes from A and the 2nd element comes from B

Binary relation

- aRb denotes that $(a,b) \in R$
- When (a,b) belongs to R , a is said to be related to b by R
- Likewise, **n-ary relations** express relationships among n elements
- Let $A_1, A_2, \dots, A_n$ be sets. An n -ary relation of these sets is a subset of $A_1 \times A_2 \times \dots \times A_n$. The sets $A_1, A_2, \dots, A_n$ are called the **domains** of the relation, and n is called its **degree**

Example

- Let A be the set of students and B be the set of courses
- Let R be the relation that consists of those pairs (a, b) where $a \in A$ and $b \in B$
- If Jason is enrolled only in CSE20, and John is enrolled in CSE20 and CSE21
- The pairs $(\text{Jason}, \text{CSE20})$, $(\text{John}, \text{CSE20})$, $(\text{John}, \text{CSE 21})$ belong to R
- But $(\text{Jason}, \text{CSE21})$ does not belong to R

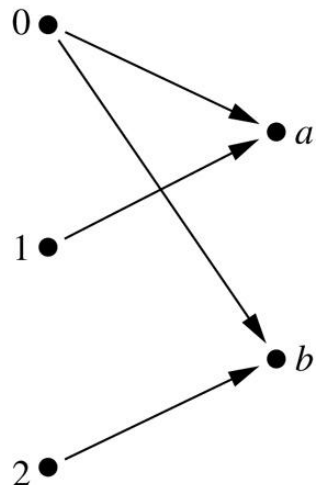
Example

- Let A be the set of all cities, and let B be the set of the 50 states in US. Define a relation R by specifying (a,b) belongs to R if city a is in state b
- For instance, (Boulder, Colorado), (Bangor, Maine), (Ann Arbor, Michigan), (Middletown, New Jersey), (Middletown, New York), (Cupertino, California), and (Red Bank, New Jersey) are in R

Example

- Let $A=\{0, 1, 2\}$ and $B=\{a, b\}$. Then $\{(0, a), (0, b), (1, a), (2, b)\}$ is a relation from A to B
- That is $0Ra$ but not $1Rb$

© The McGraw-Hill Companies, Inc. all rights reserved.



R	a	b
0	×	×
1	×	
2		×

Functions as relations

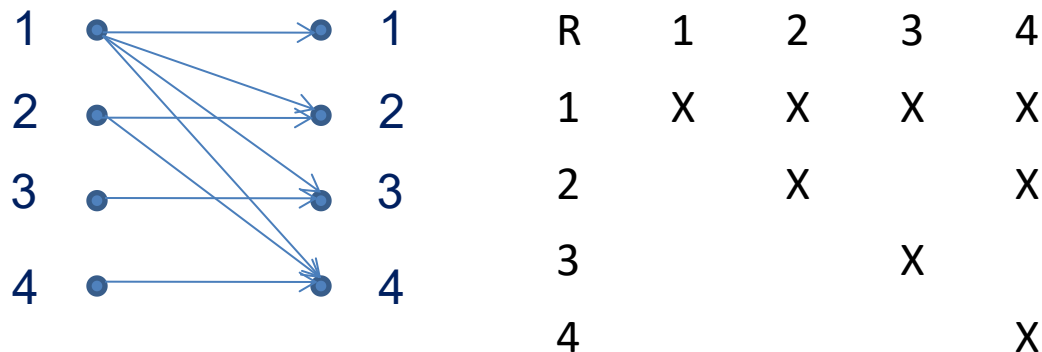
- Recall that a function f from a set A to a set B assigns **exactly one** element of B to each element of A
- The graph of f is the set of ordered pairs (a, b) such that $b=f(a)$
- Because the graph of f is a subset of $A \times B$, it is a relation from A to B
- Furthermore, the graph of a function has the property that every element of A is the first element of exactly one ordered pair of the graph

Functions as relations

- Conversely, if R is a relation from A to B such that every element in A is the first element of exactly one ordered pair of R , then a function can be defined with R as its graph
- A relation can be used to express one-to-many relationship between the elements of the sets A and B where an element of A may be related to more than one element of B
- A function represents a relation where exactly one element of B is related to each element of A
- Relations are a generalization of functions

Relation on a set

- A relation on the set A is a relation from A to A , i.e., a subset of $A \times A$
- Let A be the set $\{1, 2, 3, 4\}$. Which ordered pairs are in the relation $R = \{(a, b) \mid a \text{ divides } b\}$?
- $R = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 4), (3, 3), (4, 4)\}$



Example

- Consider these relations on set of integers

$$R_1 = \{(a,b) \mid a \leq b\}$$

$$R_2 = \{(a,b) \mid a > b\}$$

$$R_3 = \{(a,b) \mid a = b \text{ or } a = -b\}$$

$$R_4 = \{(a,b) \mid a = b\}$$

$$R_5 = \{(a,b) \mid a = b + 1\}$$

$$R_6 = \{(a,b) \mid a + b \leq 3\}$$

Which of these relations contain each of the pairs (1,1), (1,2), (2,1), (1, -1) and (2, 2)?

- (1,1) is in R_1 , R_3 , R_4 and R_6 ; (1,2) is in R_1 and R_6 ; (2,1) is in R_2 , R_5 , and R_6 ; (1,-1) is in R_2 , R_3 , and R_6 ; (2,2) is in R_1 , R_3 , and R_4

Example

- How many relations are there on a set with n elements?
- A relation on a set A is a subset of $A \times A$
- As $A \times A$ has n^2 elements, there are 2^{n^2} subsets
- Thus there are 2^{n^2} relations on a set with n elements
- That is, there are $2^{3^2} = 2^9 = 512$ relations on the set $\{a, b, c\}$

Properties of relations: Reflexive

- In some relations an element is always related to itself
- Let R be the relation on the set of all people consisting of pairs (x,y) where x and y have the same mother and the same father. Then $x R x$ for every person x
- A relation R on a set A is called **reflexive** if $(a,a) \in R$ for every element $a \in A$
- The relation R on the set A is reflexive if $\forall a((a,a) \in R)$

Example

- Consider these relations on $\{1, 2, 3, 4\}$

$$R_1 = \{(1,1), (1,2), (2,1), (2,2), (3,4), (4,1), (4,4)\}$$

$$R_2 = \{(1,1), (1,2), (2,1)\}$$

$$R_3 = \{(1,1), (1,2), (1,4), (2,1), (2,2), (3,3), (4,1), (4,4)\}$$

$$R_4 = \{(2,1), (3,1), (3,2), (4,1), (4,2), (4,3)\}$$

$$R_5 = \{(1,1), (1,2), (1,3), (1,4), (2,2), (2,3), (2,4), (3,3), (3,4), (4,4)\}$$

$$R_6 = \{(3,4)\}$$

Which of these relations are reflexive?

- R_3 and R_5 are reflexive as both contain all pairs of the (a,a)
- Is the “divides” relation on the set of positive integers reflexive?

Symmetric

- In some relations an element is related to a second element if and only if the 2nd element is also related to the 1st element
- A relation R on a set A is called **symmetric** if $(b,a) \in R$ whenever $(a,b) \in R$ for all $a, b \in A$
- The relation R on the set A is symmetric if $\forall a \forall b ((a,b) \in R \rightarrow (b,a) \in R)$
- A relation is symmetric if and only if a is related to b implies that b is related to a

Antisymmetric

- A relation R on a set A such that for all $a, b \in A$, if $(a, b) \in R$ and $(b, a) \in R$, then $a=b$ is called **antisymmetric**
- Similarly, the relation R is antisymmetric if $\forall a \forall b ((a, b) \in R \wedge (b, a) \in R) \rightarrow (a=b)$
- A relation is **antisymmetric** if and only if there are no pairs of distinct elements a and b with a related to b and b related to a
- That is, the only way to have a related to b and b related to a is for a and b to be the same element

Symmetric and antisymmetric

- The terms symmetric and antisymmetric are not opposites as a relation can have both of these properties or may lack both of them
- A relation cannot be both symmetric and antisymmetric if it contains some pair of the form (a, b) where $a \neq b$

Example

- Consider these relations on $\{1, 2, 3, 4\}$

$$R_1 = \{(1,1), (1,2), (2,1), (2,2), (3,4), (4,1), (4,4)\}$$

$$R_2 = \{(1,1), (1,2), (2,1)\}$$

$$R_3 = \{(1,1), (1,2), (1,4), (2,1), (2,2), (3,3), (4,1), (4,4)\}$$

$$R_4 = \{(2,1), (3,1), (3,2), (4,1), (4,2), (4,3)\}$$

$$R_5 = \{(1,1), (1,2), (1,3), (1,4), (2,2), (2,3), (2,4), (3,3), (3,4), (4,4)\}$$

$$R_6 = \{(3,4)\}$$

Which of these relations are symmetric or antisymmetric?

- R_2 and R_3 are symmetric: each $(a,b) \rightarrow (b,a)$ in the relation
- R_4 , R_5 , and R_6 are all antisymmetric: no pair of elements a and b with $a \neq b$ s.t. (a, b) and (b, a) are both in the relation

Example

- Which are symmetric and antisymmetric

$$R_1 = \{(a, b) \mid a \leq b\}$$

$$R_2 = \{(a, b) \mid a > b\}$$

$$R_3 = \{(a, b) \mid a = b \text{ or } a = -b\}$$

$$R_4 = \{(a, b) \mid a = b\}$$

$$R_5 = \{(a, b) \mid a = b + 1\}$$

$$R_6 = \{(a, b) \mid a + b \leq 3\}$$

- Symmetric: R_3, R_4, R_6 . R_3 is symmetric, if $a = b$ (or $a = -b$), then $b = a$ ($b = -a$), R_4 is symmetric as $a = b$ implies $b = a$, R_6 is symmetric as $a + b \leq 3$ implies $b + a \leq 3$
- Antisymmetric: R_1, R_2, R_4, R_5 . R_1 is antisymmetric as $a \leq b$ and $b \leq a$ imply $a = b$. R_2 is antisymmetric as it is impossible to have $a > b$ and $b > a$, R_4 is antisymmetric as two elements are related w.r.t. R_4 if and only if they are equal. R_5 is antisymmetric as it is impossible to have $a = b + 1$ and $b = a + 1$

Transitive

- A relation R on a set A is called **transitive** if whenever $(a,b) \in R$ and $(b,c) \in R$ then $(a,c) \in R$ for all $a, b, c \in A$
- Using quantifiers, we see that a relation R is **transitive** if we have
$$\forall a \forall b \forall c ((a,b) \in R \wedge (b,c) \in R) \rightarrow (a,c) \in R$$

Example

- Which one is transitive?

$$R_1 = \{(1,1), (1,2), (2,1), (2,2), (3,4), (4,1), (4,4)\}$$

$$R_2 = \{(1,1), (1,2), (2,1)\}$$

$$R_3 = \{(1,1), (1,2), (1,4), (2,1), (2,2), (3,3), (4,1), (4,4)\}$$

$$R_4 = \{(2,1), (3,1), (3,2), (4,1), (4,2), (4,3)\}$$

$$R_5 = \{(1,1), (1,2), (1,3), (1,4), (2,2), (2,3), (2,4), (3,3), (3,4), (4,4)\}$$

$$R_6 = \{(3,4)\}$$

- R_4 R_5 R_6 are transitive
- R_1 is not transitive as $(3,1)$ is not in R_1
- R_2 is not transitive as $(2,2)$ is not in R_2
- R_3 is not transitive as $(4,2)$ is not in R_3

Example

- Which are symmetric and antisymmetric

$$R_1 = \{(a,b) \mid a \leq b\}$$

$$R_2 = \{(a,b) \mid a > b\}$$

$$R_3 = \{(a,b) \mid a = b \text{ or } a = -b\}$$

$$R_4 = \{(a,b) \mid a = b\}$$

$$R_5 = \{(a,b) \mid a = b + 1\}$$

$$R_6 = \{(a,b) \mid a + b \leq 3\}$$

- R_1 is transitive as $a \leq b$ and $b \leq c$ implies $a \leq c$. R_2 is transitive
- R_3 , R_4 are transitive
- R_5 is not transitive (e.g., $(2,1)$, $(1,0)$). R_6 is not transitive (e.g. $(2,1)$, $(1,2)$)

Combining relations

- Relations from A to B are subsets of $A \times B$, two relations can be combined in any way that two sets can be combined
- Let $A=\{1,2,3\}$ and $B=\{1,2,3,4\}$. The relations $R_1=\{(1,1),(2,2),(3,3)\}$ and $R_2=\{(1,1),(1,2),(1,3),(1,4)\}$ can be combined
- $R_1 \cup R_2 = \{(1,1),(1,2),(1,3),(1,4),(2,2),(2,3),(3,3)\}$
- $R_1 \cap R_2 = \{(1,1)\}$
- $R_1 - R_2 = \{(2,2),(3,3)\}$
- $R_2 - R_1 = \{(1,2),(1,3),(1,4)\}$

Example

- Let $R_1 = \{(x,y) \mid x < y\}$ and $R_2 = \{(x,y) \mid x > y\}$. What are $R_1 \cup R_2$, $R_1 \cap R_2$, $R_1 - R_2$, $R_2 - R_1$, and $R_1 \oplus R_2$?
- **Symmetric difference** of A and B: denoted by $A \oplus B$, is the set containing those elements in either A or B, but not in both A and B
- We note that $(x,y) \in R_1 \cup R_2$, if and only if $(x,y) \in R_1$ or $(x,y) \in R_2$, it follows that $R_1 \cup R_2 = \{(x,y) \mid x \neq y\}$
- Likewise, $R_1 \cap R_2 = \emptyset$, $R_1 - R_2 = R_1$, $R_2 - R_1 = R_2$,
 $R_1 \oplus R_2 = R_1 \cup R_2 - R_1 \cap R_2 = \{(x,y) \mid x \neq y\}$

Composite of relations

- Let R be a relation from a set A to a set B , and S a relation from B to a set C .
- The **composite** of R and S is the relation consisting of ordered pairs (a,c) where $a \in A$, $c \in C$ and for which there exists an element $b \in B$ s.t. $(a,b) \in R$ and $(b,c) \in S$. We denote the **composite of R and S** by $S \circ R$
- Need to find the 2nd element of ordered pair in R the same as the 1st element of ordered pair in S

Example

- What is the composite of the relations R and S , where R is the relation from $\{1,2,3\}$ to $\{1,2,3,4\}$ with $R=\{(1,1),(1,4),(2,3),(3,1),(3,4)\}$ and S is the relation from $\{1,2,3,4\}$ to $\{0,1,2\}$ with $S=\{(1,0),(2,0),(3,1),(3,2),(4,1)\}$?
- Need to find the 2nd element in the ordered pair in R is the same as the 1st element of order pair in S
- $S \circ R = \{(1,0),(1,1),(2,1),(2,2),(3,0),(3,1)\}$

Power of relation

- Let R be a relation on the set A . The **powers** $R^n, n=1,2,3,\dots$, are defined recursively by $R^1=R$,
 $R^{n+1}=R^n \circ R$
- Example: Let $R=\{(1,1),(2,1),(3,2),(4,3)\}$. Find the powers $R^n, n=2,3,4,\dots$
- $R^2=R \circ R$, we find $R^2=\{(1,1),(2,1),(3,1),(4,2)\}$,
 $R^3=R^2 \circ R, R^3=\{(1,1),(2,1),(3,1),(4,1)\}$,
 $R^4=\{(1,1),(2,1),(3,1),(4,1)\}$.
- It also follows $R^n=R^3$ for $n=5,6,7,\dots$

Transitive

- Theorem: The relation R on a set A is **transitive** if and only if $R^n \subseteq R$
- Proof: We first prove the “if” part. Suppose $R^n \subseteq R$ for $n=1,2,3,\dots$. In particular $R^2 \subseteq R$. To see this implies R is transitive, note that if $(a,b) \in R$, and $(b,c) \in R$, then by definition of composition $(a,c) \in R^2$. Because $R^2 \subseteq R$, this means that $(a,c) \in R$. Hence R is transitive

Transitive

- We will use mathematical induction to prove the “only if” part
- Note $n=1$, the theorem is trivially true
- Assume that $R^n \subseteq R$, where n is a positive integer. This is the induction hypothesis. To complete the inductive step, we must show that this implies that R^{n+1} is also a subset of R
- To show this, assume that $(a,b) \in R^{n+1}$. Because $R^{n+1} = R^n \circ R$, there is an element x with $x \in A$ s.t. $(a,x) \in R$, and $(x,b) \in R^n$. The inductive hypothesis, i.e., $R^n \subseteq R$, implies that $(x,b) \in R$. As R is transitive, and $(a,x) \in R$, and $(x,b) \in R$, it follows that $(a,b) \in R$. This shows that $R^{n+1} \subseteq R$, completing the proof